



Fig. 3: ProMetric I imaging colorimeter
(Credit: www.radiantvisionsystems.com)

preventing additional investments from being made into flawed assemblies. Unlike human inspection, Radiant automated visual inspection system can find and size significant small defects instantaneously and repeatedly, regardless of their location.

Blemish detection module. Imatest's Blemish Detect module measures quality, uniformity and defects. It comes in GUI-based versions for laboratory use and in non-GUI-based version for manufacturing testing. The method can be applied to any type of flat-screen display including smartphone, tablet, computer monitor and large-screen TV. The module can test for hot or dead pixels, visible bright and dark blemishes, luminance and colour uniformity, and gamma response. An additional measurement like colour and tonal response can be added as needed.

Colorimeter. When measuring brightness levels and contrast, a colorimeter is used. It measures light from the display through three filters. It measures light and colour similar to the way a human eye would—filters are red, green and blue. Since colorimeter requires only three filters for testing contrast ratios, there is less noise when measuring black, so accuracy of the black level is higher than with a spectrophotometer.

A colorimeter provides consistent, accurate and repeatable data to support the high degree of precision needed for quality, for all types of display technologies, production lines or manufacturing locations.

ProMetric I Imaging has National Institute of Standards and Technology (NIST)-traceable four-colour factory

calibration. Paired with a spectrometer, ProMetric I camera can continue to perform four-colour calibrations so that device monitoring accuracy remains consistent with display quality specifications on the line. If it has several line changes, ProMetric I can auto-calibrate to new colour thresholds for each new line of displays without removing the camera system from the line for re-calibration.

X-Rite i1Display Pro colorimeter is another example.

Ultra-high precision spectroradiometer-based system. This system is designed to measure the spectral power distribution of a source. It measures spectral signatures of objects from a distance, which can vary from a few metres to many kilometres. Measurements can be done either radiometrically to obtain radiance as a function of time or spectrally, yielding radiance as a function of wavelength.

A spectrophotometer works in a similar way to a colorimeter, but instead of reading light in three bands, it has 31 filters to measure the entire colour spectrum. This helps it measure a wider range of colours with more accuracy.

However, the disadvantage is less accurate readings on black. X-Rite i1Pro 2 spectrophotometer can be used for colour readings including grayscale, saturation and Gretag Macbeth test.

Online monitor calibration

Calibrating a monitor means measuring the response of the monitor and then compensating for non-ideal behaviour elsewhere. For example,



Fig. 4: X-Rite i1Pro 2 spectrophotometer
(Credit: www.anandtech.com)

in the video card driver, the user is supposed to change the properties of the monitor itself to let it approach the ideal condition better.

There are a number of free monitor testing materials available in the market. Just choose a test and read the instructions to learn how images should appear for calibration. You can test the display, contrast, clock and phase, gamma calibration, sharpness, white saturation, black level, inversion, gradient, response time, viewing angle, contrast ratio and subpixel layout. There is both an offline test that can be downloaded and used on a computer without an Internet connection, and an online test where you can access these monitor testing tools online. There are a number of online monitor tests available on Lagom.nl.

One of the best-known monitor testing programs is Passmark's MonitorTest software that gives a full-screen view of various tests. It works with multiple monitor setups and all resolutions. It supports looped testing and over 30 different pattern tests.

Another monitor testing program is DisplayMate. Then, there are testers that come with video card drivers as free software, like NVIDIA's GeForce for high-resolution display testing.

SpectraCal's CalMAN suite is a powerful set of tools. It utilises a custom workflow to measure brightness, contrast, gamma, grayscale, saturation, gamut and Gretag Macbeth. It can also be used to calibrate a display and measure its accuracy.

To sum up

To improve supply chain performance, consumer electronics manufacturing companies rely on display test and measurement solutions. This helps reduce production costs and ensures high-quality end products. Automated visual inspection systems accommodate multiple screen technologies and sizes, including OLED, LCD, projection and more. For a fully-automated quality control for display, systems can be implemented throughout the manufacturing process. Makers of smartphones, tablets and other display devices rely on testing to ensure a perfect end-user experience. **EFY**